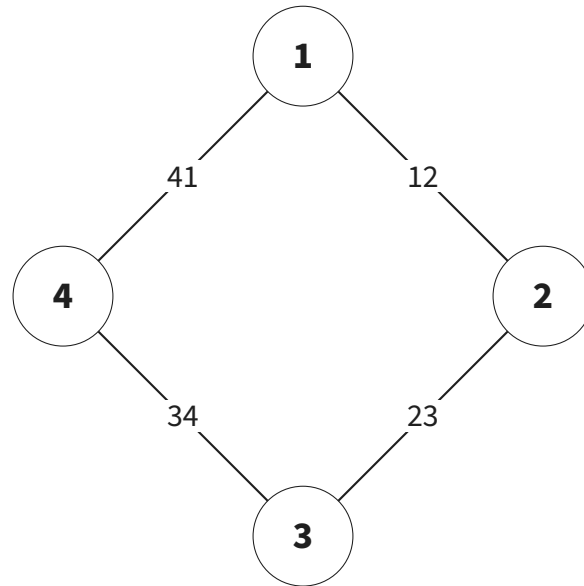


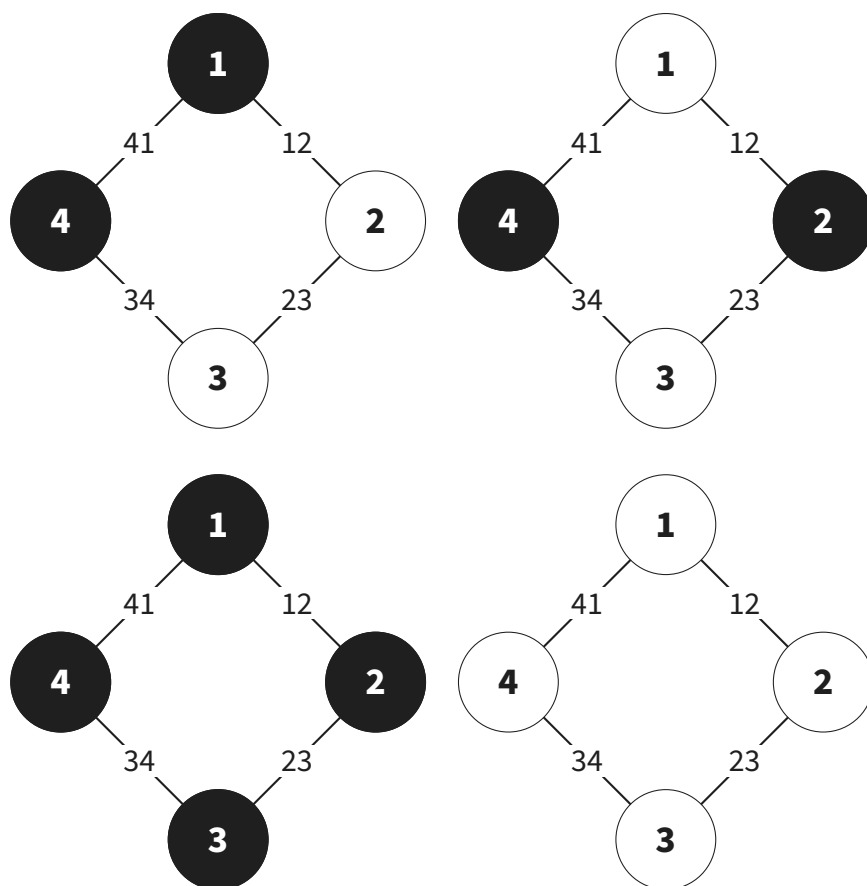
Problem 1: Put a two-sided counter on each lamp. White is OFF; dark is ON. Start all OFF. Choose a line and flip both counters at its ends. Flip that same pair again. Try another line and its undo.



Problem 2: Start all OFF each time. Make only lamps 1 and 2 ON. Then make only lamps 1 and 3 ON. Point to the lines you used.

Draw one attempt you want to remember.

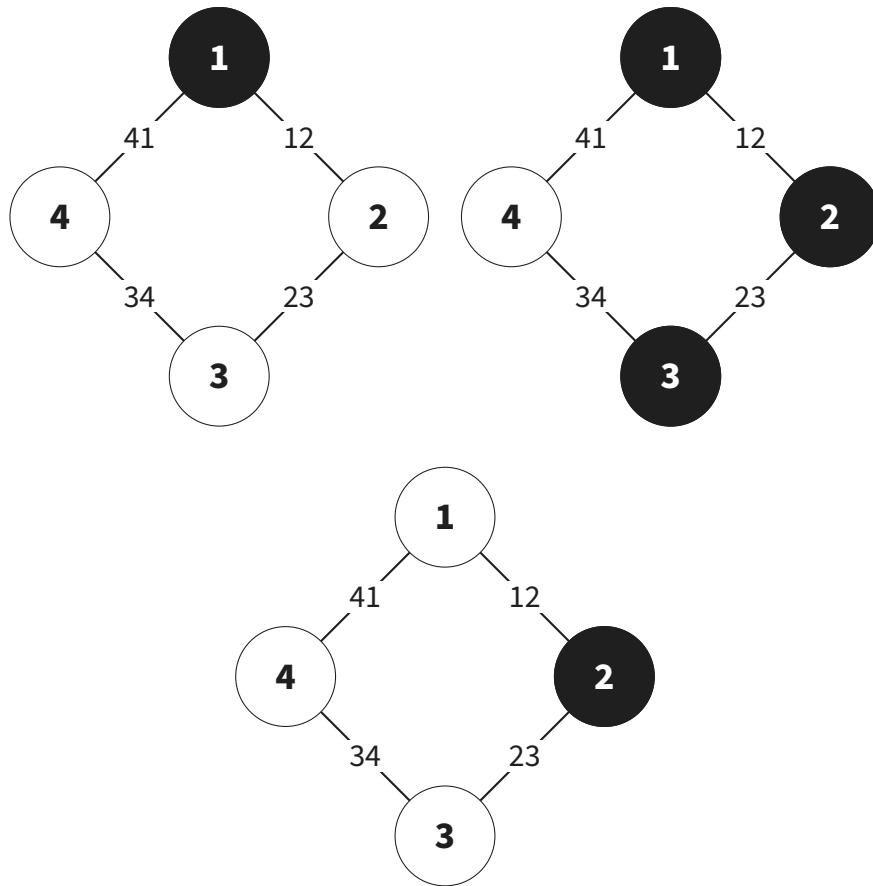
Problem 3: Start all OFF for each picture. Flip the two ends of a line each move. Make the dark lamps ON and the white lamps OFF. Mark a picture when you make it.



Problem 4: Choose a picture you made. Make it again. Undo your moves to return to all OFF. Show someone your moves in both directions.

Draw or mark the picture you chose.

Problem 5: Start all OFF. Try to make each picture using the two-end rule. Circle any picture you make. Leave the others uncircled for now.



Problem 6: Choose one uncircled picture. Change the rule: you may also flip just lamp 1. Try your picture again. Show which move uses the new rule.

Draw the picture you made with the new rule.